

Smart Parking Platform

Project Kickoff Document

Prepared by: \$oftware Corp - Software Development Studio
Document: Vision & Scope + Software Requirements Specification (SRS)
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Vision and Scope

Project Acquisition

The acquisition for this project happened on the basis of a client demand-driven approach rather than on a market-driven basis. A regional consortium comprising a municipality transportation department and some private parking facilities owners issued a Request for Proposals (RFP) for one platform for modernizing drivers' search for parking spots in garages, booking and payment process for those spots, and monitoring and pricing their facility by operators.

\$Software Corp's development team presented a proposal for this RFP based on its past experience in developing for mobility services sector. The proposal started with analysis of pain points of the consortium – constant downtown traffic due to circling of drivers looking for parking spots, and lack of real-time data on facility occupancy for operators – and not with a traditional sales presentation. The contract was awarded after a competitive tender and discussion of the scope.

Acquisition summary: Client-driven, through RFP / competitive tender. Revenue is linked to milestone-based delivery with a defined scope and timeframe, i.e. it is project-based (not product).

Company Introduction

\$Software Corp is a hypothetical software development agency working with theoretically unlimited amounts of engineering capacity and funding. The company focuses on developing end-to-end, cloud-based platforms to connect common customers with the owners of physical infrastructure – from a disparate array of offline activities to one cohesive set of digital products.

Not being a regular software agency, \$Software Corp follows an outlined go-to-market strategy where the company specializes in a few verticals, proves its competence publicly and wins customers via an inbound research-led approach. One of such verticals is urban mobility and smart-city infrastructure. Thus, a Smart Parking Platform is a natural product for the studio.

Project Overview

The Smart Parking Platform is an application that will enable users to find, reserve and pay for parking spots in parking lots that are connected to the service. It is delivered using both the web portal and the mobile app, and allows parking availability, navigation and management services to both parking spots' users and parking lot operators.

The engagement is a project: to develop a plan and manage the creation of the system mentioned above – gathering of requirements, working with stakeholders, planning, budgeting, risk and team management. The deliverable is the developed product for a particular customer.

Business Need

People who need parking spots waste a lot of time looking for them, which results in additional traffic and wasted time and productivity. At the same time, parking lot operators do not have enough instruments to control their lots' occupancy rates and pricing.

Vision Statement

The Smart Parking Platform is an online parking solution for drivers and parking operators who lose valuable time and money through inefficient, offline parking. This system brings together real-time parking space availability, guaranteed parking reservations, payments, and operators' dashboard

Project Scope

In Scope

- The ability to register users and authenticate them as drivers, operators, or administrators.
- The real-time parking-space availability of the participating parking facilities.
- The interactive map showing available parking facilities.
- Parking reservation functionality with confirmation.
- Digital payments via third-party payment providers.
- History of reservation and receipt of the same.
- The operators' dashboard of parking facilities (facilities management, pricing, and occupancy).
- Occupancy analytics.
- Notification and alert functionality for the users (alerts regarding expiration, available parking spaces).
- Integration with the external map / navigation services.

Out of Scope

- The manufacture and installation of parking facilities' hardware (gates, sensors, LPR cameras); the platform uses it but doesn't make it.
- On-street metered parking regulation and citations issuing.
- Monthly parking lease contract management.
- Payment processing and payment institution activity (handled by third party).
- Valet and autonomous vehicle management.

4. Research: Existing Software Landscape

Prior to developing the specific details of the platform, we conducted a competitive analysis of the market for existing parking software to determine strengths, weaknesses, and areas where there is room for differentiation in the offering.

4.1 Driver-Facing Consumer Apps

This type of application is used by drivers to find, reserve and pay for parking spaces. Major players in the market include ParkMobile, SpotHero, ParkWhiz, PayByPhone, and BestParking. ParkMobile focuses on on-street metered parking and municipal parking lot payments and integrates with the city systems of hundreds of cities in the USA. SpotHero and ParkWhiz provide for pre-reservation of off-street parking spaces in private garages and lots usually with a price much lower than the drive-in price. PayByPhone provides pay-per-use meter services and serves as the official parking app in many cities.

Weakness common to all of these apps is that their inventory is made up almost exclusively of commercial garages and lots, and a common issue in the reviews is reservation being ignored

by the parking attendant who either does not know about the reservation or reports that the garage is already full.

4.2 Operator-Facing Management Platforms

In the supply chain, management platforms like AirGarage, FLASH, T2 Systems, Parking BOXX, and SkiData offer operators a cloud-based control panel to manage a parking facility: set prices, control access credentials, handle transactions and report on revenue and occupancy. Contemporary solutions in this segment offer license plate recognition (LPR), real-time occupancy and monitoring, enforcement, and dynamic pricing that adjusts the price based on occupancy levels.

Weakness of these platforms is that they are designed to serve operators and property owners rather than drivers and therefore do not offer drivers any unified view of occupancy and available spaces.

4.3 Competitive Comparison

4.4 Shortcomings and Opportunities

- There is no platform that brings together live driver-facing availability with the true occupancy data of the operator, which is at the heart of the "reservation not honored" issue.
- Driver apps and operator platforms are two different products, hence, the prices, availability, and bookings aren't always in sync.
- Each operator optimizes independently for their lot; there is no neutral, cross-operator marketplace that offers live spots for drivers.

4.5 What Sets Our Platform Apart

- Two-sided in nature: One platform catering to the needs of both drivers and operators, using a single source of the truth for space availability.
- Real-time, space-based availability linked to the actual occupancy data of the operator, thereby ensuring a live spot for every booking made.
- One-stop shop for reservations, payments, invoices, and navigation for the driver as well as occupancy reports and pricing for the operator.
- Cross operator neutral marketplace: Drivers compare and book from multiple operators; operators get demand that wouldn't be possible otherwise.

Software Requirements Specification (SRS)

5.1 Introduction

5.1.1 Purpose

The purpose of this document is to define the software requirements for the Smart Parking Platform: web and mobile application allowing users to find, reserve, and pay for parking at garage parking in real time as well as to monitor and manage parking lots for the operators.

5.1.2 Scope

The product encompasses: registration and authentication, real-time availability, a map, reservation system, online payments, transactions history, operator panel, occupancy statistics, notifications, and integration with mapping and payments services. Manufacturing of hardware devices, enforcement of regulations on-street parking, and providing payments service are explicitly out of scope (see section 3.4).

3.2.1 Users

- Drivers / mobile application users - search, book, pay, navigate and manage vehicles and receipts.
- Parking facility managers - manage parking facilities and slots, define pricing policies and monitor occupancy rates.
- Finance and billing department - handle payments, reconciliations and generation of income reports.
- System Administrators - manage users, roles and configurations of the system.
- City transportation departments - consume aggregated/occupancy reports according to agreement.

3.2.2 System Environment

System is offered as native mobile applications (iOS and Android) and web applications that operate using cloud-hosted services and APIs. System incorporates services from third parties for navigation, payment processing and real time feeds from IoT occupancy sensors and camera/LPR devices.

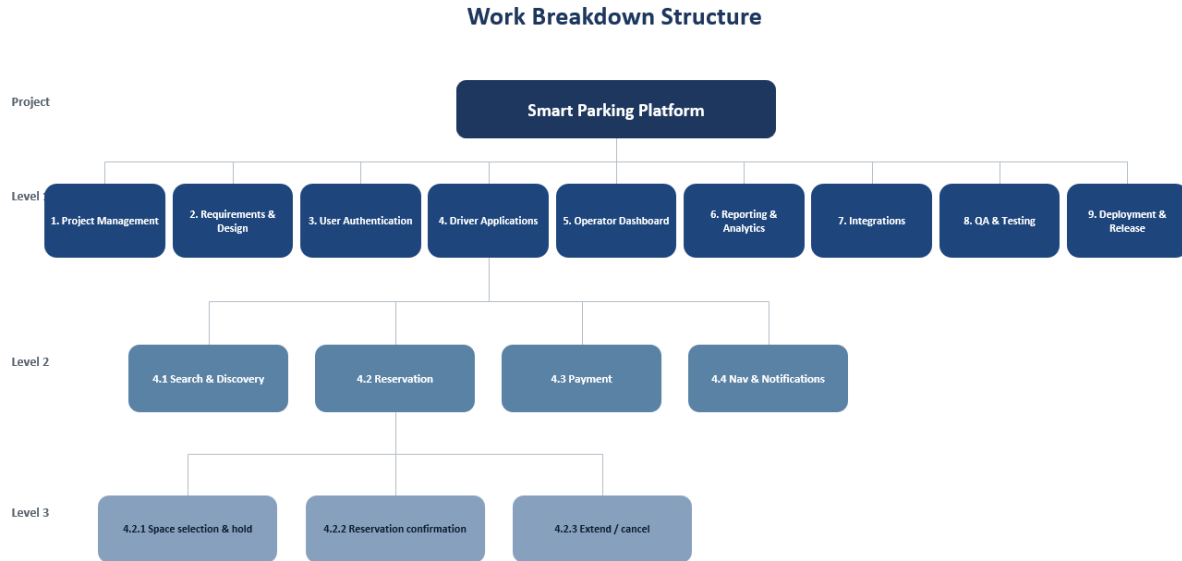
3.2.3 Constraints

- The solution should incorporate services from third parties and depend on their availability.
- The solution should be compliant with PCI-DSS and all other privacy related legal requirements.
- The solution should be designed and implemented within semester-based timeframe.

3.2.4 Assumptions & Dependencies

- Users and devices are connected to Internet or Intranet.
- Park facilities have occupancy sensors capabilities or capacity information.
- Third party providers of payment, mapping and sensors have APIs available and defined.

Work Breakdown Structure (WBS)



Complete WBS Outline

1 Project Management

1.1 Planning

- 1.1.1 Project charter & kickoff
- 1.1.2 Vision & scope document
- 1.1.3 WBS & schedule

1.2 Monitoring & Control

- 1.2.1 Status reporting
- 1.2.2 Risk management

2 Requirements & Design

2.1 Requirements

- 2.1.1 Requirements elicitation
- 2.1.2 SRS document
- 2.1.3 Use-case modeling

2.2 Design

- 2.2.1 System architecture
- 2.2.2 UI/UX design
- 2.2.3 Database design
- 2.2.4 API design

3 User Authentication & Account Management

3.1 Registration

- 3.1.1 Email / SSO signup
- 3.1.2 Account verification

3.2 Login & Session

- 3.2.1 Secure login
- 3.2.2 Multi-factor authentication
- 3.2.3 Session management

3.3 Profile Management

- 3.3.1 User profile
- 3.3.2 Vehicle profiles

4 Driver Applications (Mobile & Web)

4.1 Parking Search & Discovery

- 4.1.1 Location search & filters
- 4.1.2 Interactive map view
- 4.1.3 Real-time availability display

4.2 Reservation

- 4.2.1 Space selection & hold
- 4.2.2 Reservation confirmation
- 4.2.3 Extend/cancel

4.3 Payment

- 4.3.1 Payment method management
- 4.3.2 Payment processing
- 4.3.3 Digital pass & receipt

4.4 Navigation & Notifications

- 4.4.1 Turn-by-turn navigation
- 4.4.2 Alerts & reminders

5 Operator Dashboard & Garage Monitoring

5.1 Facility Management

- 5.1.1 Add/edit/remove facility
- 5.1.2 Space & zone configuration

5.2 Occupancy Monitoring

- 5.2.1 Live occupancy dashboard
- 5.2.2 Sensor / camera data integration

5.3 Pricing

- 5.3.1 Base-rate configuration
- 5.3.2 Dynamic pricing rules

6 Reporting & Analytics

6.1 Occupancy & Utilization Reports

6.2 Financial Reports

- 6.2.1 Revenue reporting
- 6.2.2 Export (PDF / CSV)

6.3 Administration

- 6.3.1 User & role management
- 6.3.2 Audit logging

7 Integrations

7.1 Payment Gateway Integration

7.2 Mapping / Navigation Integration

7.3 IoT Sensor / LPR Integration

8 Quality Assurance & Testing

8.1 Test Planning

8.2 Functional & Integration Testing

8.3 Security & Performance Testing

8.4 User Acceptance Testing

9 Deployment & Release

9.1 Environment Setup

9.2 Beta Release

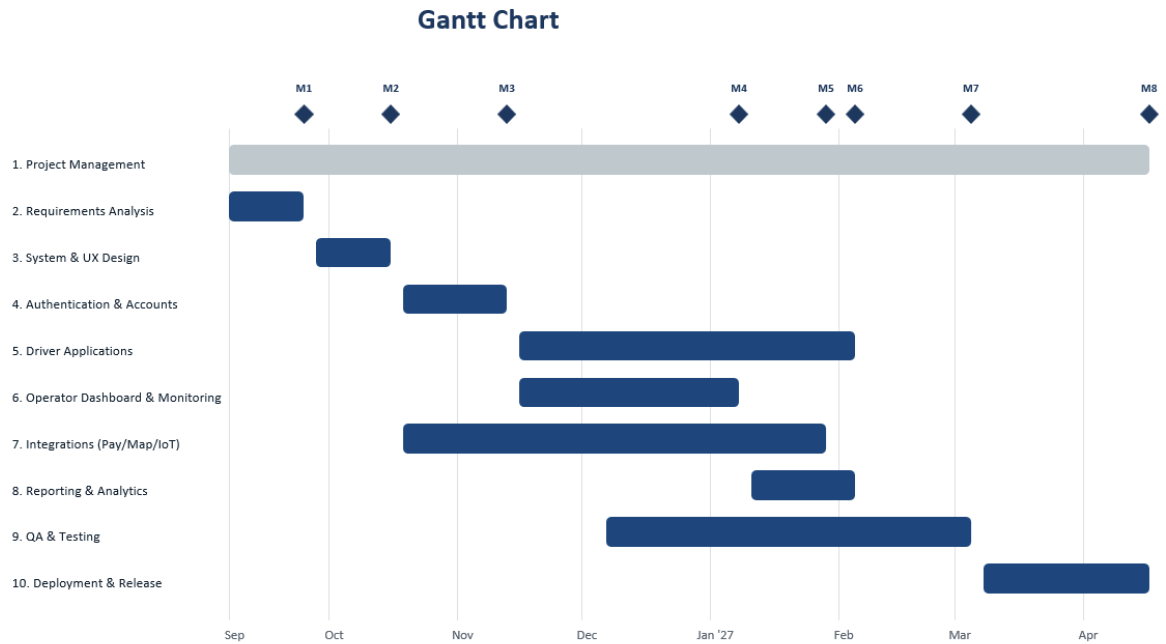
9.3 Production Release

9.4 Handoff & Documentation

Project Schedule & Timeline

Task / Phase	Start	End	Duration	Pred.	Dep.
1. Project Management	Sep 7, 2026	Apr 17, 2027	32 wks		Spans project
2. Requirements Analysis	Sep 7, 2026	Sep 25, 2026	3 wks	Kickoff	FS
3. System & UX Design	Sep 28, 2026	Oct 16, 2026	3 wks	2	FS
4. Authentication & Accounts	Oct 19, 2026	Nov 13, 2026	4 wks	3	FS
5. Driver Applications	Nov 16, 2026	Feb 5, 2027	12 wks	4	FS
6. Operator Dashboard & Monitoring	Nov 16, 2026	Jan 8, 2027	8 wks	4	SS
7. Integrations	Oct 19, 2026	Jan 29, 2027	15 wks	3	FS
8. Reporting & Analytics	Jan 11, 2027	Feb 5, 2027	4 wks	6	FS
9. QA & Testing	Dec 7, 2026	Mar 5, 2027	13 wks	5	SS
10. Deployment & Release	Mar 8, 2027	Apr 17, 2027	6 wks	9	FS

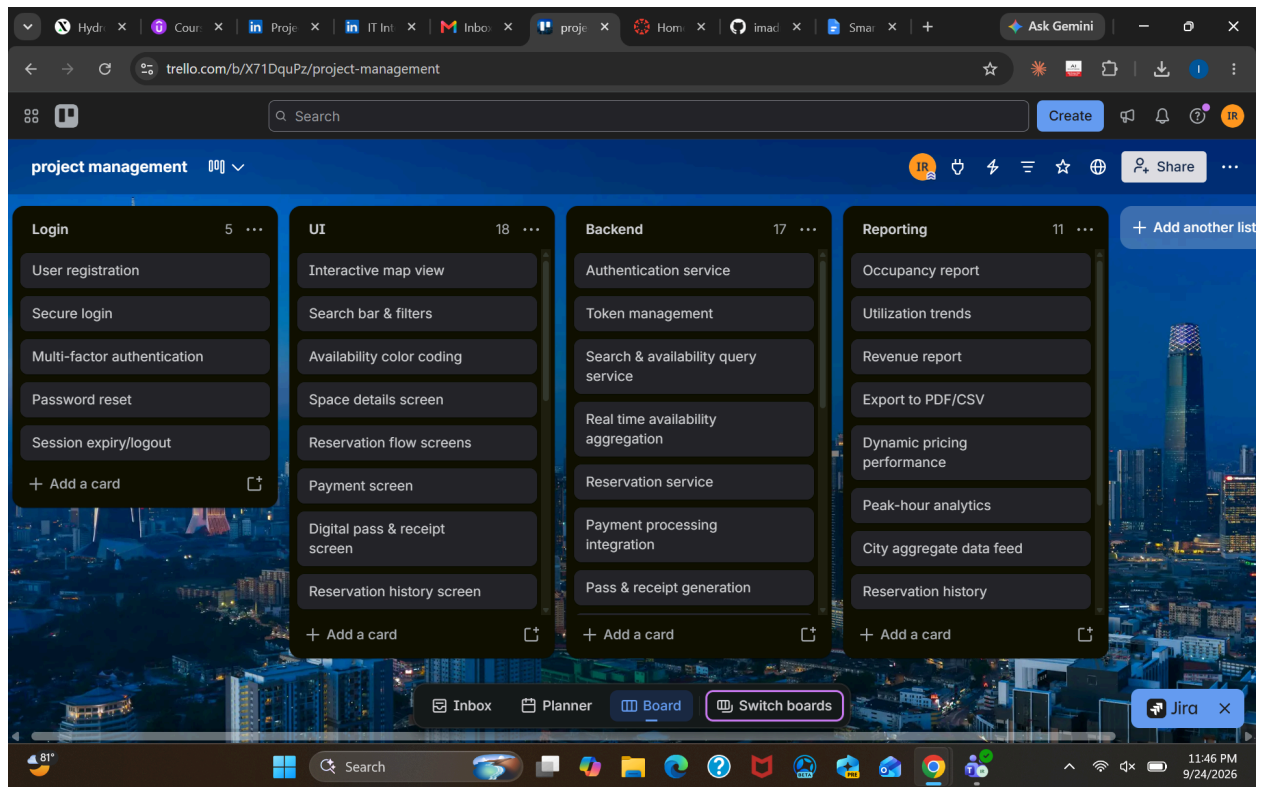
GANTT Chart



Milestones

ID	Milestone	Target Date
M1	Requirements baseline	Sep 25, 2026
M2	Design complete	Oct 16, 2026
M3	Authentication module complete	Nov 13, 2026
M4	Operator dashboard complete	Jan 8, 2027
M5	Integrations complete	Jan 29, 2027
M6	Driver app MVP	Feb 5, 2027
M7	Beta release	Mar 5, 2027
M8	Production release/handoff	Apr 17, 2027

Trello Board



Link for full: <https://trello.com/b/X71DquPz/project-management>